
Amazon ElastiCache – Complete Overview

What is Amazon ElastiCache?

♦ **Amazon ElastiCache** is a **fully managed in-memory caching service** provided by AWS that supports:

-  **Redis** (highly popular, feature-rich)
-  **Memcached** (simple, multi-threaded)

It improves performance by **storing frequently accessed data in memory**, reducing the need to access slower disk-based databases.

Why Use ElastiCache?

Feature	Benefit
 Lightning-fast access	Sub-millisecond latency
 Offloads backend DB	Reduces load on RDS, DynamoDB, etc.
 Auto-scaling (Redis Cluster Mode)	Handles millions of requests/sec
 Fully managed	No need to manage patching or setup
 Secure	Encryption, VPC, IAM integration



Cost-efficient

Reduces DB calls, improves performance



Supported Engines

Engine	Description	Use Case
 Redis	Advanced features like Pub/Sub, Streams, Persistence	Leaderboards, Caching, Real-time analytics
 Memcached	Lightweight, multi-threaded, simple key-value	Simple cache, short-lived data



Key Use Cases

1.  **In-Memory Cache**
 - Cache DB query results or API responses
2.  **Session Store**
 - Manage user sessions in stateless web apps
3.  **Leaderboards**
 - Real-time ranking systems in gaming
4.  **Real-Time Analytics**
 - Fast data access for dashboards or metrics
5.  **Pub/Sub Messaging**

- Build lightweight messaging systems (Redis only)

Core Features

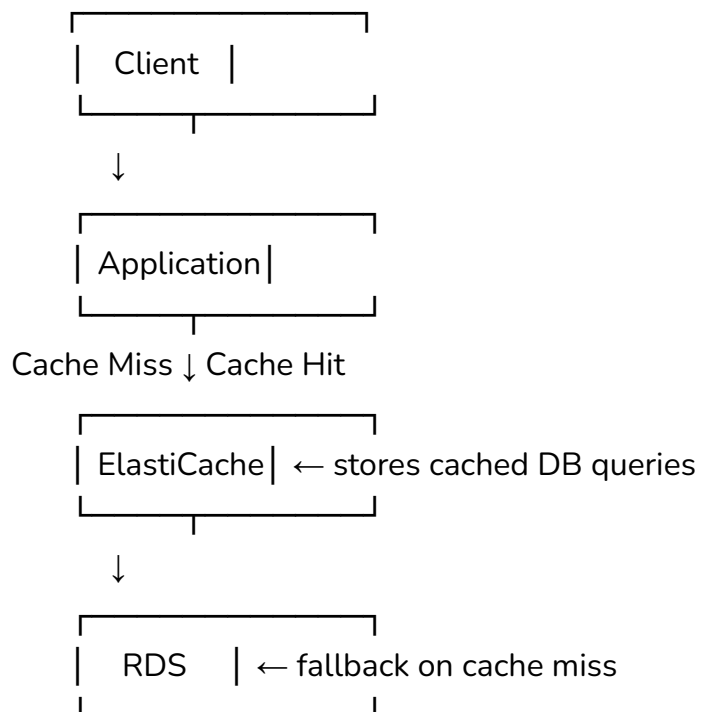
Feature	Description
 In-memory store	Sub-millisecond read/write
 Fully managed	AWS handles provisioning, patching, failover
 Multi-AZ support	High availability for Redis with automatic failover
 Monitoring	CloudWatch integration
 Security	VPC, IAM, KMS, in-transit & at-rest encryption
 Clustering	Scale horizontally across multiple shards (Redis Cluster Mode)
 Backup & Restore	Snapshot backups for Redis (not Memcached)

Security in ElastiCache

Feature 	Redis	Memcached
VPC support	✓	✓
IAM for management	✓	✓
Encryption (in-transit, at-rest)	✓	✗

Redis AUTH token	✓	✗
TLS support	✓	✗

Architecture Example: Caching for RDS



Redis vs Memcached – Quick Comparison

Feature	Redis 	Memcached 
Data types	Strings, Lists, Sets, Hashes, Streams	Strings only
Persistence	 Yes (snapshot & AOF)	 No
Clustering	 Yes (Redis Cluster)	 Limited
TLS Encryption	 Yes	 No
Multi-threaded	 No	 Yes
Pub/Sub	 Yes	 No

Pricing Overview

You pay for:

-  Node type and count (memory size & compute)
-  Data transfer
-  Backups (for Redis)
-  Multi-AZ (if enabled)

Use the [AWS Pricing Calculator](#) for precise cost estimation.

Hands-on Lab Idea: ElastiCache Redis Cache for RDS

Goal:

Use Redis to cache product data from RDS to reduce DB load.

Steps:

1. Create Redis cluster in ElastiCache
2. Launch EC2 instance (in same VPC)
3. Install Redis CLI: `sudo apt install redis-tools`

Connect to Redis and set keys:

```
redis-cli -h <redis-endpoint>
```

```
SET product:101 '{"id":101,"name":"Shoes"}' EX 600
```

```
GET product:101
```

- 4.
5. Integrate with sample Node.js or Python app

Summary

Feature	Description
What is it?	Managed Redis/Memcached caching
Benefits	Low latency, offload DB, scale reads
Engine options	Redis (advanced), Memcached (simple)

Security	VPC, IAM, Encryption
Use cases	Caching, sessions, pub/sub, analytics
Integration	Works with EC2, Lambda, RDS, API Gateway, etc.
